

Rahul Agarwal

AI Engineer | Generative AI & LLMs | Agentic AI | Computer Vision | Python

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PROFESSIONAL SUMMARY

AI Engineer with **2.7+ years** of professional experience building **Generative AI, LLM, RAG, Agentic AI**, and **Computer Vision** applications with scalable Python backends and REST APIs. Strong hands-on expertise in **Python, FastAPI, Django, Flask, Node.js**, and **SQL**. Experienced in integrating machine learning, object detection, OCR, and AI agents into real-world IIoT and enterprise solutions, and currently building **AI SaaS products** for SEO, content generation, and marketing automation.

EXPERIENCE

Aureate Labs

Feb 2026 – Present

AI Specialist

Surat, Gujarat (On-site)

- Building **AI-powered SaaS products** for SEO, AI content generation, marketing automation, and ad analytics using **LLMs**
- Developing Generative AI features, cold email automation, and campaign automation with **Node.js, TypeScript, React.js**, and **GCP**

Xconics

Apr 2025 – Feb 2026

Python & AI/ML Engineer

Mumbai, Maharashtra

- Built **FastAPI** backends for **Computer Vision** and **Industrial IoT** systems for BCCL, Mahindra Logistics, and PCBL
- Deployed **object detection, face recognition**, and **ANPR** models with MQTT, PostgreSQL, and AWS S3 for real-time alerts

Avinya Digital Innovations

Jan 2024 – Mar 2025

Python Developer & AI/ML Engineer

Surat, Gujarat

- Developed Python backends and automation tools integrating **Generative AI, LLMs**, and computer vision
- Built **LLM-powered** web scraping and job-automation products used across 10+ platforms

elInfochips (An Arrow Company)

Jul 2023 – Aug 2023

Summer Intern – ASIC Flow Orientation

Ahmedabad, Gujarat

SKILLS

Generative AI & LLMs: LLMs, RAG, Agentic AI, AI Agents, Prompt Engineering, LLM API Integration, AI Content Generation

Machine Learning: Machine Learning, Deep Learning, NLP, Model Integration

Computer Vision: Object Detection, YOLO, OpenCV, OCR, ANPR, Face Recognition, MediaPipe, Image Processing

Languages & Frameworks: Python, TypeScript, SQL, FastAPI, Django, Flask, Node.js, React.js, Streamlit, REST APIs

Databases: PostgreSQL, MongoDB, Firebase

Cloud & IoT: GCP, AWS S3, MQTT, IIoT

Automation & Tools: Instantly, Groq, Web Scraping, Workflow Automation, SEO Automation

PROJECTS

AI Ad & E-commerce Analytics Platform

(Aureate Labs)

Generative AI Marketing Analytics & Optimization SaaS

- Building a **unified dashboard** that brings advertising and e-commerce data together for performance tracking
- Adding **LLM-powered insights**, AI ad creative generation, campaign planning, and automated optimization

AI SEO & Content Generation Platforms

(Aureate Labs)

Two SaaS Products in the Space of Ahrefs, Semrush, SurferSEO, and Outrank

- Developing SEO tools for **keyword research**, **content analysis**, and ranking optimization
- Building **LLM-based content generation pipelines** that produce SEO-optimized articles at scale

Cold Email Automation System

(Aureate Labs)

AI-Driven Outbound Sales Automation

- Building automated cold email workflows using **Instantly** and integrated tools for campaign setup, sequencing, and follow-ups

Coal Mine Driver Safety & Vehicle Identification System

(Xconics)

Computer Vision Backend – Face Recognition & ANPR

(BCCL)

- Built a Python backend using **computer vision** for driver face recognition and unsafe driving detection in coal mines
- Implemented number plate recognition (**ANPR**) to automatically log truck entry details via REST APIs

AI Voice Agent

Agentic AI & LLM Voice Automation

(HR Consultants - US)

- Designed and developed a Python backend integrating **LLM APIs** for automated HR voice communication workflows
- Implemented APIs, **prompt engineering**, and response processing for conversational AI

AI Studio

Generative AI Image Generation Backend

(Textile)

- Developed a Python backend to generate AI-driven photoshoot and flat-lay clothing visuals using **Generative AI** models
- Exposed AI pipelines through **REST APIs** for frontend consumption

Forklift Monitoring System

(Xconics)

Industrial IoT Backend for Productivity Tracking

(Mahindra Logistics)

- Designed a **FastAPI** backend using **MQTT** and **PostgreSQL** to track forklift engine status, load activity, and zone movement

Anti-Theft Surveillance System

(Xconics)

Real-Time Object Detection Backend

(PCBL)

- Integrated **object detection** models with FastAPI for real-time alert generation
- Handled image processing and storage using **AWS S3** via backend APIs

Hand Movement Detection System

(Xconics)

IMU Sensor-Based Motion Analysis for Industrial Harvesting

- Built a **motion tracking system** using IMU sensors to detect hand gestures and identify optimal plucking movements
- Provided efficiency insights, **reducing wasted movement** and improving operational accuracy

AI Web Scraper

(Avinya Digital Innovations)

LLM-Powered Data Extraction Tool

- Built a flexible Python scraper using **Crawl** and **Groq** with a **Streamlit** UI for user input
- Supports multiple output formats and scrapes **10+ platforms** at ~10s per page

Upwork Extensions

(Avinya Digital Innovations)

Job Automation & Generative AI Cover Letter Tool

- Created a browser extension and website for instant job alerts and **auto-application in under 2 minutes**
- Used **Generative AI** to write personalized cover letters and attach media based on the user's profile

Exercise Pose Recognition & Set Counting(MSP Cup 2022)

Computer Vision Pose Estimation – R-10 (Asia-Pacific) Level Competition

- Developed using **Python**, **MediaPipe**, and joint-angle calculations
- Features include timer, rep counter, exercise guidance, pose detection, and error detection

Drone Delivery System(SSIP Grant)

Research and Feasibility Study for Last-Mile Delivery

- Researched the feasibility of drone-based last-mile delivery, evaluating cost-effectiveness, safety, and real-world constraints

EDUCATION

B.E. – Electronics and Communication Engineering2020 – 2024

Sarvajanik College of Engineering and Technology (ECE Dept., NBA Accredited)Surat, Gujarat

- CGPA:** 8.29 | **Minor:** Artificial Intelligence and Machine Learning (CGPA: 8.66)

ACHIEVEMENTS

1st Position – IEEE Multimedia Signal Processing Cup 2022Aug 2022

R-10 (Asia-Pacific) level competition organized by IEEE SPS Gujarat Section

SSIP Grant of Rs. 1 Lakh – Drone Delivery System (SKYBOUND)2024

Startup Support and Innovation Program (SSIP), Government of Gujarat

CERTIFICATIONS

100 Days of Code: The Complete Python Pro Bootcamp

Udemy – Dr. Angela Yu52 hours